THANGARAJ T

AI/ML ENGINEER | PYTHON FULL-STACK DEVELOPER

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GitHub: github.com/tmthangaraj1980-AIML | Portfolio: tmthangaraj1980-aiml.github.io/AI-ML-Portfolio/

PROFESSIONAL SUMMARY

AI/ML engineer focused on building practical intelligent applications across machine learning, deep learning, NLP, computer vision and generative AI. Hands-on experience developing retrieval-augmented generation (RAG) systems, semantic search, LLM-based assistants and LangGraph-orchestrated AI-agent workflows. Combines Python model development with FastAPI REST services and HTML/CSS/JavaScript interfaces to deliver end-to-end AI solutions. Experienced in applied document intelligence, OCR, model evaluation and Git-based project delivery.

FEATURED AI PROJECT | MEDIGUIDE AI

Healthcare Article-Aware RAG Assistant | FastAPI, LangGraph, Groq, FAISS

Developed a responsive healthcare learning interface with an embedded AI assistant using vanilla JavaScript and a FastAPI backend. Indexed 100 MedlinePlus topics with Sentence Transformers embeddings and FAISS; implemented article-scoped retrieval and source-grounded answers. Used LangGraph orchestration with Groq LLM integration and insufficient-evidence handling. Recorded a local demo; not cloud deployed.

Code: github.com/tmthangaraj1980-AIML/MediGuide-AI

TECHNICAL SKILLS

AI / Machine Learning: scikit-learn, classification, regression, clustering, feature engineering, evaluation

Deep Learning / Vision: TensorFlow/Keras, PyTorch, CNNs, OpenCV, YOLO, image classification

Generative AI / AI Agents: LLMs, LangGraph, LangChain, RAG, FAISS, Groq, Gemini, Ollama, Sentence Transformers

NLP / Document AI: text preprocessing, TF-IDF, embeddings, OCR, information extraction

Python / Full Stack: Python, FastAPI, REST APIs, Pydantic, HTML, CSS, JavaScript, SQL, SQLite, React basics

Tools: Git, GitHub, VS Code, Jupyter, Pandas, NumPy, virtual environments

PROFESSIONAL EXPERIENCE

AI/ML Intern | FemtoSoft Technologies, Chennai | Sep 2026 – Present

Developing ClearCustoms AI for customs-document extraction, validation and ICEGATE field mapping using Python, OCR/NLP, LLMs and FastAPI.

AI/ML Developer Intern | OneYes Infotech, Chennai | Aug – Sep 2026

Worked on face-recognition attendance workflows using MTCNN and OpenCV, alongside document-generation utilities.

AI/ML Intern | DataMites | Jan – Jul 2026

Built a PDF RAG chatbot and completed applied machine learning projects involving prediction, NLP and geospatial analysis.

EDUCATION

M.E. Computer Science (AI & ML) — RVS College of Engineering and Technology (in progress)

B.E. Electronics & Communication Engineering — Sri Sairam College of Engineering, Chennai